

Symplectic Deep Equilibrium Models: Volume-Preserving Fixed-Point Solvers for Multistable Implicit Representations

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Abstract

The stability guarantees that make deep equilibrium models (DEQs) usable—global contraction or monotonicity—force the implicit layer to have a *unique* fixed point. By the Banach fixed-point theorem this uniqueness is not incidental: a globally contractive map cannot host two coexisting attractors. Any system whose correct representation is genuinely *multistable*—two stable basins separated by a barrier—therefore cannot be modelled by a contractive DEQ without collapsing its basins into one. We name this tension **basin collapse** and propose **Symplectic DEQ (S-DEQ)**, which replaces the contractive forward update with a volume-preserving symplectic integrator of a learned Hamiltonian. Volume preservation removes the global contraction that forbids multistability while retaining a well-defined notion of equilibrium for implicit differentiation. We are deliberate about what this does and does not buy: abandoning contraction means convergence is *no longer guaranteed a priori*, so rather than *assuming* a Lipschitz constant $L < 1$ we *measure* the effective operator norm at the solved fixed point by Jacobian power iteration and report it. **[FILL: main multistability + convergence results]**. As a cautionary application we examine a stability-margin readout for protein fold-switching; an external leakage audit [Companion, 2027] finds the margin to be a *geometric (RMSD) confound shortcut* rather than a thermodynamic signal, establishing that representational multistability does not by itself yield predictive biological signal. We release code and report the solver’s convergence limits transparently.

1 Introduction

Deep equilibrium models [Bai et al., 2019] replace a deep stack of layers with the fixed point of a single weight-tied map $f_\theta(\cdot; x)$, solving $z^* = f_\theta(z^*; x)$ by root finding and differentiating through z^* implicitly, at constant memory. Their central practical difficulty is that an arbitrary f_θ has no guarantee that a fixed point exists, is unique, or can be reached. The dominant remedies all impose some form of *contraction*: monotone operator parameterizations [Winston & Kolter, 2020], Lipschitz or Jacobian regularization [Bai et al., 2021], and architectural caps on the forward operator norm. Each restores a unique, reliably reachable equilibrium.

This paper starts from an uncomfortable observation: *the very property that stabilizes a DEQ also forbids it from representing a multistable system*. If $f_\theta(\cdot; x)$ is a global contraction with Lipschitz constant $L < 1$, the Banach fixed-point theorem guarantees *exactly one* fixed point. A system that is genuinely bistable—two stable conformations, two competing equilibria, a hysteretic switch—has *two* stable fixed points separated by an unstable one. No contraction can express this; forced through a contractive DEQ, the two basins merge into a single averaged attractor. We call this failure mode **basin collapse**. It is not a bug in any particular implementation but a direct consequence of demanding global contraction.

The natural question is what to relax. We do not want to abandon the equilibrium formulation—the constant-memory implicit gradient is the reason to use a DEQ at all—so we keep “output = fixed point of an iterated map” and change the *map*. The geometric object that admits multiple coexisting equilibria while remaining well-behaved is a *volume-preserving* (symplectic) map: the time- h flow of a Hamiltonian system. Symplectic maps do not contract phase-space volume (their Jacobian has unit determinant), so they are not forbidden from possessing several fixed points of mixed stability type, yet they are the most structurally constrained non-contractive maps available, with a century of numerical analysis behind their integrators.

This motivates **Symplectic DEQ**: parameterize the implicit layer by a learned Hamiltonian $H_\theta(q, p; x)$ and define the forward solver as a symplectic integrator driven toward an equilibrium of H_θ .

Relaxing contraction has an honest cost, and we state it up front rather than hide it. Without global contraction there is *no a priori convergence guarantee*. We therefore treat convergence as an empirical quantity: we run the solver, and at the returned state we measure the effective Lipschitz constant $\hat{L} = \|\partial f_\theta / \partial z\|$ by Jacobian power iteration. We report $\hat{L}$ and the fixed-point residual as primary diagnostics, never a *targeted* $L < 1$ as though it were achieved. In our current implementation an operator-norm cap intended to bound the per-step Jacobian was found not to be applied inside the forward pass; preliminary power-iteration measurements indicated a non-contractive regime ($\hat{L} > 1$). We regard reliable convergence control under multistability as the central open problem of this line of work, and Section 8 is explicit about it.

Contributions.

- **Basin collapse, stated precisely.** We formalize the tension between DEQ stability guarantees and multistability: any globally contractive implicit layer has a unique fixed point (Banach) and therefore cannot represent two coexisting attractors (Proposition 1).
- **Symplectic DEQ.** An implicit layer whose forward solver is a volume-preserving symplectic integrator of a learned Hamiltonian, admitting multiple equilibria of mixed stability while retaining implicit differentiation (Section 4).
- **Measured, not assumed, convergence.** A protocol that reports the effective Lipschitz constant at the solved fixed point by Jacobian power iteration, in place of an assumed contraction (Section 4, 6).
- **An honest cautionary application.** On protein fold-switching, the naive stability-margin readout from a multistable model is shown—by an external leakage audit [Companion, 2027]—to be a geometric confound shortcut, not a thermodynamic signal (Section 7). Representational multistability is necessary but not sufficient for biological signal.

2 Related Work

Deep equilibrium models and their stabilization. DEQs [Bai et al., 2019, 2020] solve for a fixed point of a weight-tied map and backpropagate through it implicitly. Because existence, uniqueness, and convergence are not automatic, a line of work imposes contraction: monotone operator equilibrium networks [Winston & Kolter, 2020] parameterize the layer so a *unique* equilibrium is guaranteed and operator-splitting solvers provably converge; others regularize the Jacobian or bound the Lipschitz constant [Bai et al., 2021]. Our premise is that this guaranteed uniqueness is exactly what precludes multistability, and we deliberately step outside it. We are not aware of prior DEQ work whose explicit goal is to *host multiple stable equilibria* within one implicit layer.

Hamiltonian and symplectic neural networks. Hamiltonian Neural Networks [Greydanus et al., 2019] learn a scalar $H(q, p)$ and integrate Hamilton’s equations to model conservative *trajectories*; Symplectic ODE-Net [Zhong et al., 2020] adds control, and structure-preserving symplectic networks build invertible volume preservation into the architecture. These works use symplectic structure to *simulate dynamics over time*. We instead use a symplectic integrator as the *fixed-point solver of an implicit layer*: the object of interest is the equilibrium that the integration approaches, not the trajectory itself.

Closely related hybrids, and an explicit disambiguation. Two recent lines are the nearest neighbours and we differentiate from them directly. (i) Methods that embed a symplectic implicit integrator inside a fixed-point iteration for HNN training [Symplectic fixed-point HNN, 2024] share the “symplectic + fixed point” mechanics, but their target is system identification of a known dynamical system, not a learned implicit representation with designed multistability. (ii) Hamiltonian primitives that relax a learned potential to an equilibrium for energy-based generation [CHLU, 2026] share the “relax-to-equilibrium on a learned Hamiltonian” idea; we differ in keeping the DEQ implicit-gradient formulation and in centring multistability and its measurement. *Terminology caution:* the phrase “Hamiltonian DEQ” already appears in the literature with an unrelated meaning—prediction of the quantum (DFT) Hamiltonian *matrix* via a DEQ [DFT

Hamiltonian DEQ, 2024]—and has nothing to do with Hamiltonian *mechanics*; we use “symplectic DEQ” throughout to avoid the collision.

Energy landscapes and associative memory. That multiple attractors arise from an energy landscape is classical in Hopfield and dense associative memory models, and modern Hopfield networks are known to connect closely to attention and to implicit/equilibrium computation [Ramsauer et al., 2021]. That literature establishes multistability via an explicitly *dissipative* (gradient) descent on an energy. Our route is complementary and, to our knowledge, not previously taken for DEQs: multistability via a *volume-preserving* (non-dissipative) update, which is what lets us keep symplectic structure rather than enforce gradient contraction.

3 Background

Deep equilibrium layer. Given input x and a weight-tied map $f_\theta : \mathbb{R}^n \times \mathbb{R}^d \rightarrow \mathbb{R}^n$, a DEQ defines its output as a fixed point $z^\star = f_\theta(z^\star; x)$ and differentiates a loss $\ell(z^\star)$ through the implicit function theorem,

$$\frac{\partial \ell}{\partial(\cdot)} = \frac{\partial \ell}{\partial z^\star} \left(I - J_{f_\theta}(z^\star) \right)^{-1} \frac{\partial f_\theta(z^\star; x)}{\partial(\cdot)}, \quad J_{f_\theta}(z^\star) = \left. \frac{\partial f_\theta}{\partial z} \right|_{z^\star}, \quad (1)$$

without storing the forward solver’s intermediate states.

Why contraction implies uniqueness.

Proposition 1 (Contraction forbids multistability). *If $f_\theta(\cdot; x)$ is a global contraction on a complete metric space, i.e. $\|f_\theta(u; x) - f_\theta(v; x)\| \leq L\|u - v\|$ with $L < 1$, then it has exactly one fixed point, which the Picard iteration reaches from any initialization. In particular it cannot possess two stable fixed points, so a bistable target is not representable.*

This is the Banach fixed-point theorem [Banach, 1922]; we restate it only to make the design consequence explicit. Monotone/Lipschitz-bounded DEQs sit squarely in its hypotheses, which is precisely the regime we must leave to obtain multistability.

Symplectic maps and Hamiltonian flow. On phase space $(q, p) \in \mathbb{R}^{2m}$ with the canonical form $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$, a differentiable map Φ is *symplectic* if $\Phi'^\top J \Phi' = J$, which implies $\det \Phi' = 1$ (volume preservation). The flow of a Hamiltonian system $\dot{q} = \partial_p H$, $\dot{p} = -\partial_q H$ is symplectic for all time, and symplectic integrators (e.g. Störmer–Verlet/leapfrog) preserve this property at the discrete level. Crucially for us, a volume-preserving map is *not* contractive, so Proposition 1 does not apply to it—multiple equilibria of mixed stability index are not excluded.

4 Symplectic DEQ

State augmentation and learned Hamiltonian. We lift the hidden state $z \in \mathbb{R}^n$ to a phase-space pair $(q, p) \in \mathbb{R}^n \times \mathbb{R}^n$ and parameterize a learned Hamiltonian

$$H_\theta(q, p; x) = \frac{1}{2} p^\top M_\theta(x)^{-1} p + V_\theta(q; x), \quad (2)$$

with a (positive-definite) learned mass $M_\theta(x)$ and a learned potential $V_\theta(q; x)$ whose multiple minima are the basins we wish to represent. The input x is injected into both M_θ and V_θ (input injection), as in standard DEQs.

Volume-preserving forward solver. The forward pass integrates a damped Hamiltonian relaxation with a symplectic (leapfrog) step of size h , using a small, *state-independent* dissipation γ applied only to momenta

so that the *conservative* part of each update remains volume preserving:

$$p_{k+1/2} = p_k - \frac{h}{2} \nabla_q V_\theta(q_k; x), \quad (3)$$

$$q_{k+1} = q_k + h M_\theta(x)^{-1} p_{k+1/2}, \quad (4)$$

$$p_{k+1} = (1 - \gamma) \left(p_{k+1/2} - \frac{h}{2} \nabla_q V_\theta(q_{k+1}; x) \right). \quad (5)$$

The equilibria of the implicit layer are the points with $\nabla_q V_\theta = 0$ and $p = 0$; with $\gamma = 0$ the map is exactly symplectic, and γ small controls how aggressively trajectories settle into a basin *without* imposing a global contraction on the conservative directions. The DEQ fixed point we differentiate through is $z^* = q^*$ at the basin selected by the initialization/input. We emphasize that this is a *design that permits* multiple basins, not a guarantee that the solver converges to one; see below.

Implicit differentiation. At a returned equilibrium we use the standard implicit gradient (1) with f_θ the composed leapfrog update (5); in practice we use the phantom/1-step gradient approximation when the Jacobian solve is ill-conditioned, and we report which estimator is used per experiment.

Convergence is measured, not assumed. Because we have abandoned global contraction, we do *not* claim an $L < 1$ bound. Instead, at every returned fixed point we estimate the effective Lipschitz constant

$$\hat{L} = \|J_{f_\theta}(z^*)\|_2 \quad \text{by Jacobian power iteration,} \quad (6)$$

and report it alongside the fixed-point residual $\|f_\theta(z^*; x) - z^*\|$ and the iteration count. $\hat{L} < 1$ at a point certifies *local* contraction (local convergence) there; $\hat{L} \geq 1$ is reported honestly as a non-contractive solve. This diagnostic is the load-bearing measurement of the paper and is never replaced by a target value.

5 Theoretical Characterization

Observation 1 (Volume preservation permits multistability). *A volume-preserving map can possess several isolated fixed points of differing stability index. Concretely, the (damped) leapfrog map (5) for a double-well potential V_θ with two minima separated by a maximum admits, for small h, γ , two locally attracting equilibria (one per well) and one unstable equilibrium (the barrier), none of which is removed by the unit-Jacobian constraint of the conservative part.*

Proof sketch. The equilibria coincide with critical points of V_θ ; local stability follows from the eigenvalues of the linearized map at each, which for small γ inherit the index of the Hessian $\nabla^2 V_\theta$. **[FILL: full proof / exact small- h, γ stability conditions to be written and checked]** A complete statement with admissible (h, γ) ranges is deferred to Appendix A.

What we do *not* claim. We make no global convergence theorem. The honest theoretical content is asymmetric: Proposition 1 *rules out* multistability under contraction (a clean negative result), whereas Observation 1 only *permits* multistability under volume preservation (an existence statement), and says nothing about whether the learned H_θ trains to place basins where the data needs them, nor about a guaranteed basin of attraction. Those are empirical questions (Section 6) and, for guaranteed convergence, an open problem (Section 8).

6 Experiments

Reporting discipline. Every numeric cell below is a placeholder to be filled from a real run under the locked protocol of Appendix B. We do not print a number we have not measured. All metrics are reported over $S \geq 5$ seeds with bootstrap 95% confidence intervals, and the effective Lipschitz $\hat{L}$ (6) and fixed-point residual accompany every accuracy figure.

6.1 Synthetic bistable benchmark

We construct controlled double-well targets in which the ground-truth number and location of stable basins are known, and ask whether a model recovers *both* basins rather than collapsing them. Baselines: vanilla DEQ [Bai et al., 2019] and monotone DEQ [Winston & Kolter, 2020], which Proposition 1 predicts must collapse to a single averaged attractor.

Table 1: Basin recovery on the synthetic bistable benchmark. “#basins” is the number of distinct stable equilibria recovered (ground truth = 2); “barrier MAE” measures placement of the separating maximum; $\hat{L}$ is the measured operator norm at the solved fixed point. All values to be filled from a real run.

Model	#basins (GT = 2)	barrier MAE ↓	residual ↓	$\hat{L}$ (measured)
Vanilla DEQ	[FILL:]	[FILL:]	[FILL:]	[FILL:]
Monotone DEQ	[FILL:]	[FILL:]	[FILL:]	[FILL: (< 1 by construction)]
S-DEQ (ours)	[FILL:]	[FILL:]	[FILL:]	[FILL:]

6.2 Convergence diagnostics

We characterize the solver itself: residual versus iteration, the measured $\hat{L}$ distribution across inputs, and the fraction of solves that reach a fixed point within tolerance. This subsection is where the central open issue is reported rather than hidden.

Table 2: Solver convergence. “solve rate” is the fraction of inputs reaching residual $< \epsilon$ within the iteration budget; $\hat{L}$ is summarized as median [IQR]. Preliminary debugging indicated a non-contractive regime ($\hat{L} > 1$) due to an operator-norm cap not being applied in the forward pass; the definitive measured values are to be filled after the fix is verified.

Configuration	solve rate ↑	median $\hat{L}$ [IQR]	iters to tol
S-DEQ, cap <i>not</i> applied (as-found)	[FILL:]	[FILL: (prelim. > 1, non-contractive)]	[FILL:]
S-DEQ, cap applied (fixed)	[FILL:]	[FILL:]	[FILL:]

6.3 Ablations

We ablate the symplectic integrator (vs. a non-symplectic explicit step), the dissipation γ , and the mass parameterization M_θ , reporting the effect of each on basin recovery and on $\hat{L}$. [FILL: ablation table]

7 Cautionary Application: Fold-Switching Proteins

A natural application of a multistable implicit model is protein fold-switching, where one sequence adopts two distinct stable folds. The tempting claim is that a learned stability margin (e.g. derived from $\hat{L}$ or the depth of the two basins) predicts which proteins are metamorphic. **We do not make this claim, because our own audit refutes it.**

Using the companion leakage auditor [Companion, 2027], we audited the stability-margin readout against a matched structural baseline. On homology-disjoint folds the margin is *beaten* by a baseline that sees only the pairwise RMSD ($D=0.567$ vs. $C=0.876$) and retains no explanatory power once RMSD is partialled out (partial-correlation $p=0.93$). The audit verdict is SHORTCUT: the margin is a *geometric proxy*, not a thermodynamic signal. Notably the memorization split-gap test is blind here ($R=0.589 \approx D=0.567$), so a standard split-only check would have wrongly certified the margin. These audit values are reported as-is from the external audit, not as a result of S-DEQ.

The lesson is the honest one: *representational multistability is necessary but not sufficient for biological signal*. A model that can host two basins does not thereby know which proteins switch; reading a thermodynamic claim off an internal margin requires ruling out the geometric confound, and here it does not survive. We report this negative result because it is the correct boundary on what S-DEQ demonstrates.

8 Limitations

No global convergence guarantee. By design we abandon global contraction to permit multistability, so convergence to a fixed point is not guaranteed for arbitrary inputs. We report solve rate and the measured $\hat{L}$ (6) rather than asserting convergence.

Operator-norm control is the central open problem. In the as-found implementation an operator-norm cap intended to bound the per-step Jacobian was not applied inside the forward pass, and power-iteration measurements indicated a non-contractive regime ($\hat{L} > 1$). The definitive post-fix measurements are pending (Table 2); reconciling reliable convergence with genuine multistability—i.e. *local* contraction inside each basin without *global* contraction across basins—is the key unresolved question of this work.

Existence, not placement. Observation 1 permits multiple basins; it does not guarantee that training places them where the data requires, nor that the basins are well separated. We provide empirical evidence (Section 6) but no guarantee.

The protein readout is a shortcut. As shown in Section 7, the naive stability-margin readout for fold-switching is a geometric confound, not a thermodynamic signal. S-DEQ should not be presented as a fold-switch predictor on the strength of that margin.

Cost. The augmented phase-space state and the Jacobian power-iteration diagnostic add compute relative to a standard DEQ; we report wall-clock overhead in Appendix B. **[FILL: overhead numbers]**

9 Conclusion

We identified *basin collapse*: the contraction that stabilizes a DEQ also forbids it from representing a multistable system, because a global contraction has a unique fixed point. We proposed Symplectic DEQ, which trades global contraction for volume preservation—a learned Hamiltonian integrated symplectically—so that multiple equilibria of mixed stability are no longer excluded, while the implicit gradient is retained. We were explicit about the cost of this trade: convergence is no longer guaranteed, so we measure the effective Lipschitz constant at the solved fixed point instead of assuming it, and we report the regime in which our current solver is non-contractive as an open problem rather than a solved one. Finally, we drew the honest boundary on the protein application: an external audit shows the naive stability margin is a geometric shortcut, so representational multistability must not be oversold as biological signal. We release the solver and its diagnostics so that both the multistability and its limits can be reproduced.

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A Deferred Proofs

[FILL: Full statement and proof of Observation 1, including admissible (h, γ) ranges and the linearized-map eigenvalue conditions at each critical point of V_θ . Write and check before submission; do not assert a stability range that has not been verified.]

B Reproducibility and Integrity Protocol

We mirror the integrity discipline of the companion audit.

Deterministic solver. All group/seed-dependent construction is made deterministic (sorted ordering); we verify byte-identical metrics, confidence intervals, and $\hat{L}$ across two process hash seeds before reporting any number. Where deterministic values are less favourable than a seed-dependent draw, we report the deterministic values.

Measured, not targeted, Lipschitz. The effective Lipschitz constant $\hat{L}$ (6) is obtained by Jacobian power iteration at the solved fixed point and reported as measured. We never substitute a design target ($L < 1$) for a measurement, and we report non-contractive solves ($\hat{L} \geq 1$) explicitly.

No placeholder reported as a result. Numbers appear in the final manuscript only after a real run under the locked configuration; every value currently shown as **[FILL:]** is pending and must not be filled with a guess.

Release. We release an anonymized package reproducing every reported number with a single command, including the synthetic generators, the solver, and the $\hat{L}$ diagnostic. **[FILL: wall-clock and overhead figures]**